

First-Stage Summary: Key Findings

Scripts 51 & 52 — April 17, 2026

Main Findings

- **Emp-share weighting is relevant and improves on emp-weighting — for the president instrument only.** At the firm level, the presidential alignment instrument (P) rises from $F = 15$ to $F = 38$ (cycle-specific) and from $F = 5.5$ to $F = 33$ (2002-fixed) when switching from emp-weighted to emp-share-weighted. Mayor and governor instruments remain dominated by the unweighted spec ($F > 45$, $F > 100$). At the *aggregated* level, emp-share-weighted produces numerical artifacts (near-singular cells) and cannot be used.
- **Splitting the firm sample reveals no concentration: both subsamples have a strong first stage.** For BNDES extensive (cycle-specific, unweighted, coalition, pooled), the governor instrument yields $F = 64$ in top-quartile municipalities and $F = 61$ in the bottom three quartiles. The first-stage signal at the firm level is broad, not city-size-specific.
- **Splitting the *aggregated* sample reveals sharp concentration in large cities.** With $\text{muni} \times \text{year}$ FE, the governor instrument reaches $F = 23.5$ in top-quartile municipalities but only $F = 0.4$ in the rest (full-sample F was 2.2 — diluted by small cities). For party alignment the joint M+G+P combo gives $F = 14.6$ in large cities vs. $F = 1.3$ elsewhere. The sector-level first stage is viable only for top-quartile municipalities.

Firm Level: Weighting Comparison — BNDES Extensive, Coalition, Pooled-count

All firms · Coalition alignment · Pooled-count exposure

Baseline	Weighting	Instrument combo					
		M	G	P	M+G	M+P	M+G+P
Cycle-specific	Unweighted	45.5	103.2	0.9	63.3	22.9	43.4
	Emp-weighted	0.2	0.9	15.0	0.5	7.9	5.9
	Emp-share-wt'd	0.4	0.8	38.2	0.5	11.0	7.6
2002-fixed	Unweighted	13.7	23.7	24.4	16.3	20.2	22.6
	Emp-weighted	0.2	2.5	5.5	1.4	2.9	4.1
	Emp-share-wt'd	1.2	2.7	33.2	1.2	9.7	7.2

Shaded = $F > 10$. Emp-share-wt'd improves strongly on emp-weighted for P (president), but unweighted dominates for M (mayor) and G (governor). Firm + muni×year FE. SEs clustered by firm + muni.

Firm Level: Split Sample — BNDES Extensive, Coalition, Cycle-specific, Unweighted

Pooled-count exposure · Main instrument family

Sample	Instrument combo					
	M	G	P	M+G	M+P	M+G+P
All firms	45.5	103.2	0.9	63.3	22.9	43.4
Top Q4	21.6	63.9	1.3	37.1	11.3	24.9
Bottom 3Q	33.3	61.1	0.1	40.6	16.8	28.8

Shaded = $F > 10$. Both subsamples show a strong first stage. Top Q4: 37.9M obs; Bottom 3Q: 5.2M obs. Firm + muni×year FE. SEs clustered by firm + muni.

Implication: the alignment instrument is a valid first stage in large and small cities alike at the firm level. City-size heterogeneity arises only at the aggregated (sector) level (next slide).

Aggregated Level: Split Sample — BNDES Extensive, Unweighted, Pooled-count, Cycle-specific

BNDES sector · Owner-count aggregation · Exposure control

Alignment	FE	Sample	Instrument combo					
			M	G	P	M+G	M+P	M+G+P
Coalition	MxJ+JxT	Top Q4	1.2	2.6	1.6	0.8	1.0	0.7
		Bottom 3Q	0.0	0.3	0.3	0.2	0.9	0.8
	MxJ+MxT	Top Q4	1.2	23.5	1.0	16.1	0.4	13.4
		Bottom 3Q	0.1	0.4	1.5	0.2	0.7	0.5
Party	MxJ+JxT	Top Q4	1.4	2.4	3.3	1.6	6.0	14.6
		Bottom 3Q	2.9	1.4	0.0	0.8	0.7	1.3
	MxJ+MxT	Top Q4	0.3	0.0	2.3	0.2	1.2	1.1
		Bottom 3Q	0.0	0.5	2.5	0.1	1.2	0.9

Shaded = $F > 10$. Full-sample coalition MxJ+MxT: $G=2.2$, $M+G=3.0$, $M+G+P=3.3$ (all weak — diluted by small cities).

Muni×sector + year FE. SEs clustered by muni + sector.

Selected Regression Tables

Firm: BNDES Extensive — Coal · Cyc · Unweighted · All firms

	Dep. var: 1(BNDES _{fmt} > 0)					
	M	G	P	M+G	M+P	M+G+P
$FA_{coal.}^{mayor}$	0.002*** (0.000)			0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
$FA_{coal.}^{gov}$		0.003*** (0.000)		0.003*** (0.000)		0.003*** (0.000)
$FA_{coal.}^{pres}$			0.000 (0.000)		0.000 (0.000)	-0.001* (0.000)
<i>F</i> -statistic	45.5	103.2	0.9	63.3	22.9	43.4
Observations	43,184,682	43,184,682	43,184,682	43,184,682	43,184,682	43,184,682
<i>R</i> ²	0.286	0.286	0.286	0.286	0.286	0.286

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. SEs clustered by firm + muni in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm: BNDES Extensive — Coal · Cyc · Emp-share-weighted · All firms

	Dep. var: 1(BNDES _{fmt} > 0)					
	M	G	P	M+G	M+P	M+G+P
$FA_{coal.}^{mayor}$	-0.002 (0.004)			-0.003 (0.004)	-0.001 (0.004)	-0.002 (0.004)
$FA_{coal.}^{gov}$		0.003 (0.004)		0.003 (0.004)		0.006 (0.004)
$FA_{coal.}^{pres}$			-0.029*** (0.005)		-0.022*** (0.005)	-0.023*** (0.005)
<i>F</i> -statistic	0.4	0.8	38.2	0.5	11.0	7.6
Observations	23,309,976	23,309,976	23,309,976	23,309,976	23,309,976	23,309,976
<i>R</i> ²	0.555	0.551	0.551	0.555	0.555	0.555

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. SEs clustered by firm + muni in parentheses. Weighted by firm's pre-election municipality employment share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm Split: BNDES Extensive — Top-Quartile Municipalities

Cycle-specific · Unweighted · Coalition · Pooled-count

	Dep. var: 1($\text{BNDES}_{fmt} > 0$)					
	M	G	P	M+G	M+P	M+G+P
$FA_{coal.}^{mayor}$	0.002*** (0.000)			0.001*** (0.000)	0.002*** (0.000)	0.001*** (0.000)
$FA_{coal.}^{gov}$		0.003*** (0.000)		0.003*** (0.000)		0.003*** (0.000)
$FA_{coal.}^{pres}$			0.000 (0.000)		0.000 (0.000)	-0.000 (0.000)
<i>F</i> -statistic	21.6	63.9	1.3	37.1	11.3	24.9
Observations	37,924,455	37,924,455	37,924,455	37,924,455	37,924,455	37,924,455
R^2	0.287	0.287	0.287	0.287	0.287	0.287

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Sample: top-quartile municipalities by mean RAIS employment (2002–2017). SEs clustered by firm + muni in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm Split: BNDES Extensive — Bottom Three Quartiles

Cycle-specific · Unweighted · Coalition · Pooled-count

	Dep. var: 1($\text{BNDES}_{fmt} > 0$)					
	M	G	P	M+G	M+P	M+G+P
$FA_{coal.}^{mayor}$	0.003*** (0.001)			0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
$FA_{coal.}^{gov}$		0.005*** (0.001)		0.005*** (0.001)		0.005*** (0.001)
$FA_{coal.}^{pres}$			0.000 (0.001)		-0.000 (0.001)	-0.002* (0.001)
F-statistic	33.3	61.1	0.1	40.6	16.8	28.8
Observations	5,197,189	5,197,189	5,197,189	5,197,189	5,197,189	5,197,189
R^2	0.319	0.319	0.319	0.319	0.319	0.319

Firm + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Sample: bottom three quartiles by mean RAIS employment (2002–2017). SEs clustered by firm + muni in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Split: BNDES Extensive, Coalition, MxJ+MxT — Top-Quartile Municipalities

BNDES sector · Unweighted · Cycle-specific · Pooled-count

	Dep. var: Share of firms receiving a BNDES loan _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{coal.}^{mayor}$	-0.003 (0.003)			-0.001 (0.003)	0.000 (0.003)	-0.001 (0.003)
$\overline{FA}_{coal.}^{gov}$		0.012** (0.002)		0.013** (0.002)		0.013** (0.002)
$\overline{FA}_{coal.}^{pres}$			-0.016 (0.016)		-0.012 (0.014)	-0.012 (0.014)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	1.2	23.5	1.0	16.1	0.4	13.4
Observations	63,205	54,330	54,330	53,311	53,311	53,311
R ²	0.665	0.697	0.697	0.708	0.708	0.708

Muni × sector + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Owner-count aggregation. Unweighted regressions. Sample: top-quartile municipalities by mean RAIS employment (2002–2017). Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Agg Split: BNDES Extensive, Coalition, MxJ+MxT — Bottom Three Quartiles

BNDES sector · Unweighted · Cycle-specific · Pooled-count

	Dep. var: Share of firms receiving a BNDES loan _{jmt}					
	M	G	P	M+G	M+P	M+G+P
$\overline{FA}_{coal.}^{mayor}$	-0.001 (0.002)			-0.001 (0.002)	-0.001 (0.003)	-0.001 (0.003)
$\overline{FA}_{coal.}^{gov}$		0.002 (0.003)		0.001 (0.004)		0.001 (0.004)
$\overline{FA}_{coal.}^{pres}$			-0.009 (0.008)		-0.007 (0.010)	-0.007 (0.010)
Exposure Control	Yes	Yes	Yes	Yes	Yes	Yes
F-statistic	0.1	0.4	1.5	0.2	0.7	0.5
Observations	112,383	96,303	96,303	90,512	90,512	90,512
R ²	0.607	0.635	0.635	0.645	0.645	0.645

Muni × sector + muni × year FE. Coalition alignment. Cycle-specific baseline. Pooled-count exposure. Owner-count aggregation. Unweighted regressions. Sample: bottom three quartiles by mean RAIS employment (2002–2017). Exposure controls interacted with year. SEs clustered by muni + BNDES sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.